



KrishokChat: A Safety-First Agentic Agricultural Advisory System for Bangladeshi Smallholder Farmers

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CAPSTONE
2026
NSU · ECE

85,979

benchmark instances, 4 tracks

1.9×

Bengali QA over best zero-shot

95–97%

on-device crop diagnosis

<30 ms

offline Android inference

2 paisa

marginal cost per AI answer

16123

human escalation, always free

The Problem

Bangladesh has 47 million smallholder farming households with no reliable advisory in their own language. A wrong pesticide dose can harm a family and pollute local water. Standard chatbots always give a chemical dosage answer when asked, whether or not the answer is correct. Crop disease left undetected costs 20–40% of yield.

Our Approach: A Layered Trust Stack

We built a system where every answer is tied back to an official government publication. The pipeline runs in strict order:

Publications → **Knowledge Graph** → **Safety Check** → **Hybrid Retrieval** → **Bengali Generation** → **Dosage Verifier** → **Cited Answer or Human Escalation**

Each stage was added because our measurements showed the previous stage was not enough on its own. See column 4 for the failure data behind each decision.

Key Contributions

- 85,979-instance Bengali agricultural benchmark across 4 evaluation tracks, every instance citation-traced to source pages.
- 2,882-node agricultural knowledge graph from 284 official publications across 13 national research institutions.
- Register-aware hybrid retrieval that closes the colloquial/formal language gap measured at R@10 0.970 (formal) vs. 0.093 (colloquial).
- Pre-retrieval safety gate for 6 agrochemical risk categories with a 14-field dosage verifier; only 0.31% non-compliance on 323 unsafe queries.
- On-device crop disease models (1.6–21.8 MB) achieving 95–97% top-1 accuracy across 5 crops, with no internet required.
- 722-image soil-moisture dataset with paired tensiometer readings; first field-collected soil dataset for Bangladesh agricultural AI.

Real-World Impact

Health. Any query involving a banned chemical, overdose risk, or self-harm is blocked and routed to the 16123 Krishi Call Center or 999. This safety layer is present in every tier of the product at no cost to the farmer.

Environment. Correct dosing based on verified sources reduces over-application. Bangladesh sensor-irrigation pilots report up to 63% water savings and 50% urea savings, pointing in the same direction as our soil work.

Access. Works in standard Bengali and 6 regional dialects, accepts voice input, and runs fully offline on mid-range Android phones.

Roadmap

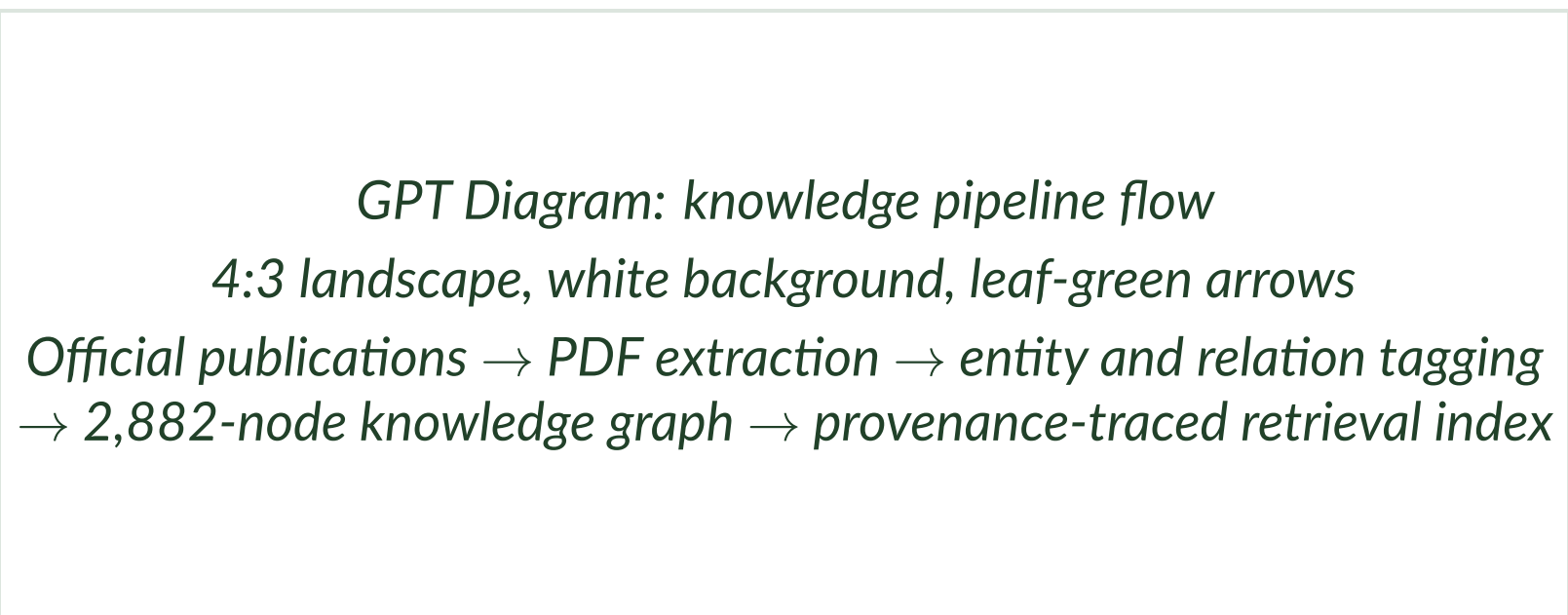
Voice output for low-literacy users · broader dialect support · soil-vision refinement for irrigation advisories · district-level pilot with a DAE field extension office.

Research Assets: What We Built and Released

Asset	Scale	What makes it reliable
Benchmark (4 tracks)	85,979 stances	General QA 28,993 · Treatment 11,224 (66.3% chemical) · Table 25,650 · Safety 20,112. Every instance linked to a source page.
Knowledge graph	2,882 nodes, 17,501 triples	19,768 entities from 284 publications at 13 institutions (BARC, BARI, BRRI, DAE and others).
Farmer-language queries	1,000 queries	300 face-to-face interviews in Rajshahi and Natore, capturing 6 regional dialects.
Soil-moisture dataset	722 field images	Paired tensiometer readings (0.0 to 21.5 kPa) across 14 crops, 46 series with strict train/test separation. Pabna, May 2026.

All four datasets released CC-BY-4.0 on Hugging Face.

From Official Publications to Retrieval Nodes



Collected in the Field, Not Scraped from the Web

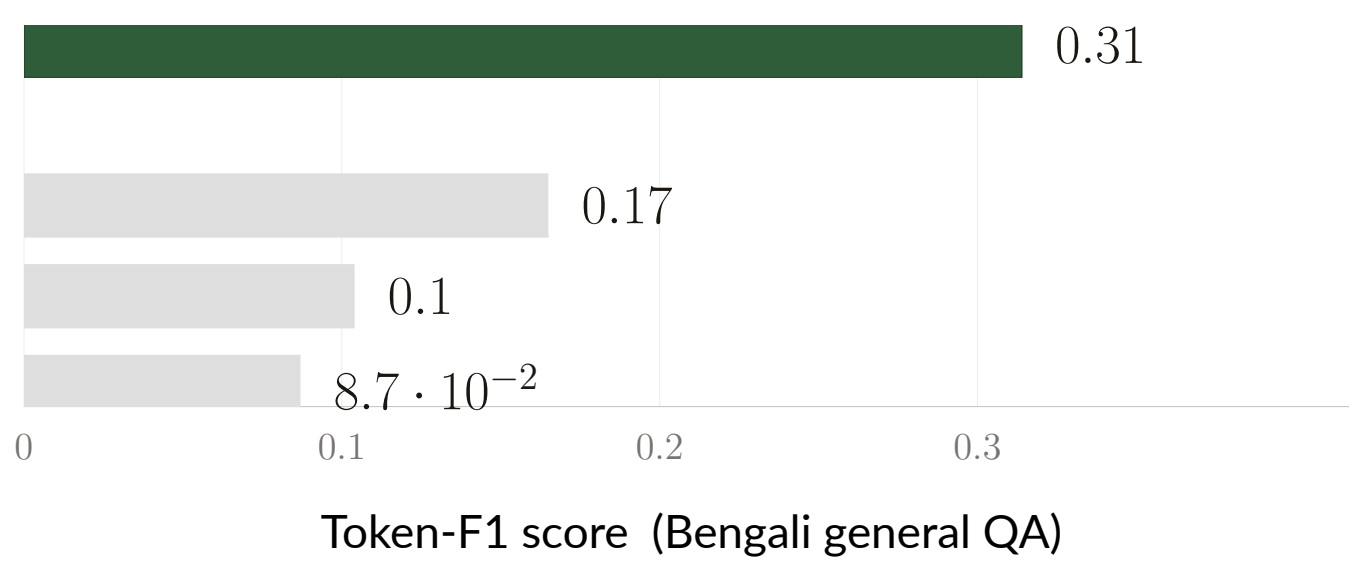


0.0 kPa (wet)

8.0 kPa (dry)

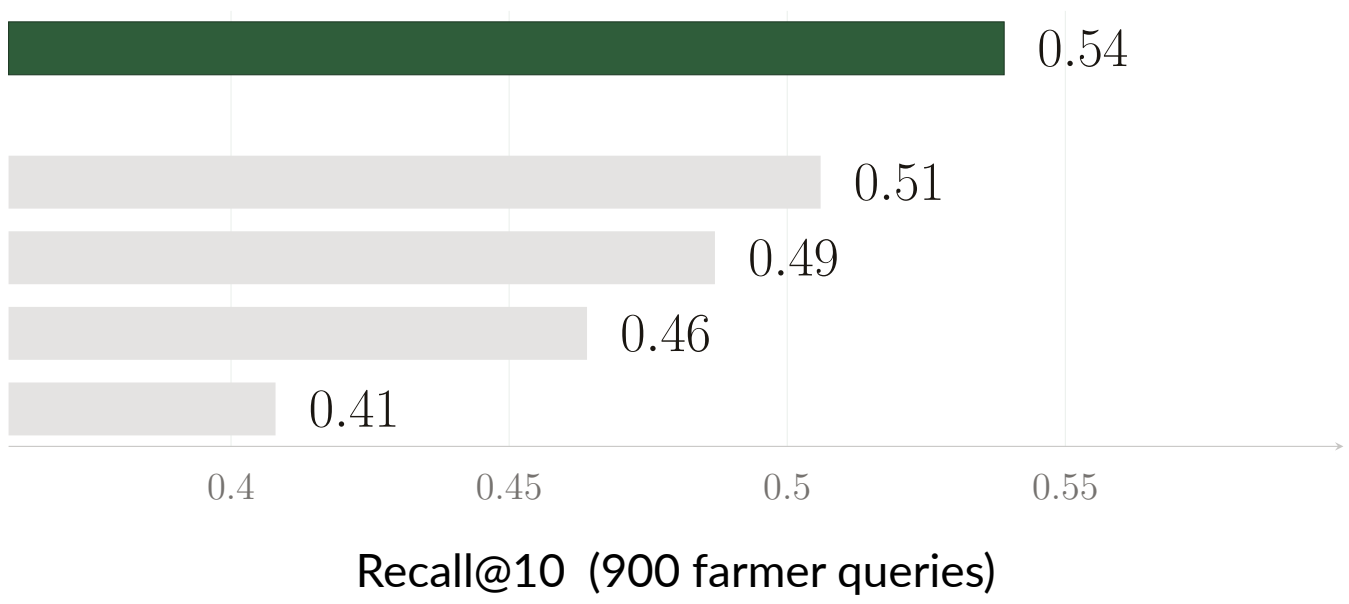
- Each soil image is paired with a tensiometer reading taken at the same time, giving a physical measurement instead of an annotator label.
- The 46 measurement series are split by series boundary so that near-duplicate images cannot appear in both training and test sets.
- Every benchmark instance links back to a specific page in an official publication, making answers auditable.

Language Model: Fine-Tuned 4B Outperforms Larger Models



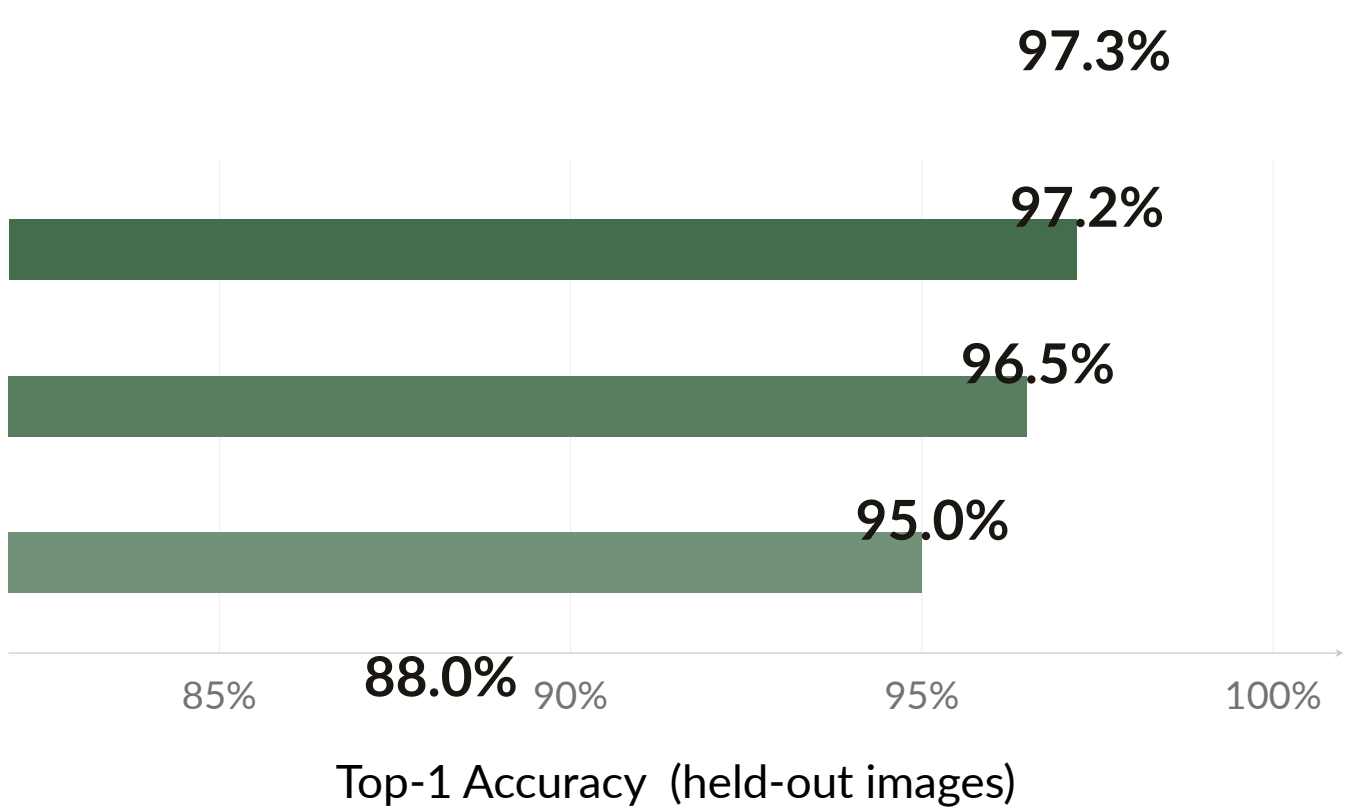
1.9× gain over best zero-shot model ($p \approx 6.7 \times 10^{-5}$)
LoRA fine-tune of Gemma-4-E4B, 4-bit quantised for on-device deployment.

Retrieval: Hybrid RRF Fusion Wins Across All Methods



+9% over dense retrieval ($p < 0.001$, Wilcoxon signed-rank). Formal queries: R@10 = 0.970. Colloquial: R@10 drops to 0.093 (dense alone).

On-Device Crop Disease Accuracy (Top-1)



436/437 Bengali treatment verified (99.8%)
<30 ms on-device inference mid-range Android

Safety Gate: 99.7% of Unsafe Queries Blocked

322 blocked (99.7%) 1

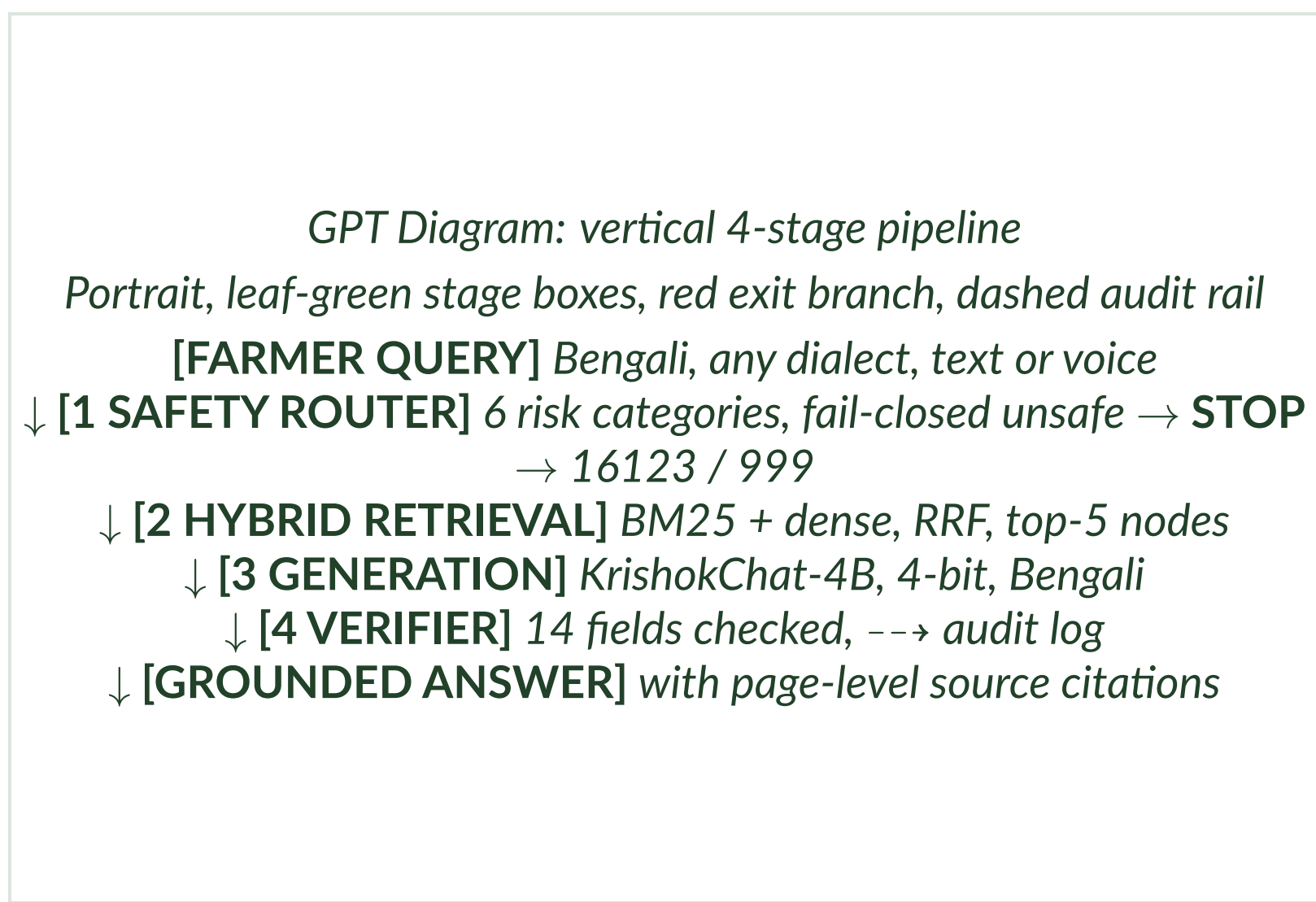
Out of 323 unsafe test queries, only 1 passed through. When evidence is absent from the knowledge base, the system withholds its answer rather than guessing.

Evaluation: benchmark held-out split, 900 farmer queries, per-crop held-out images. All models run on-device; no API calls.

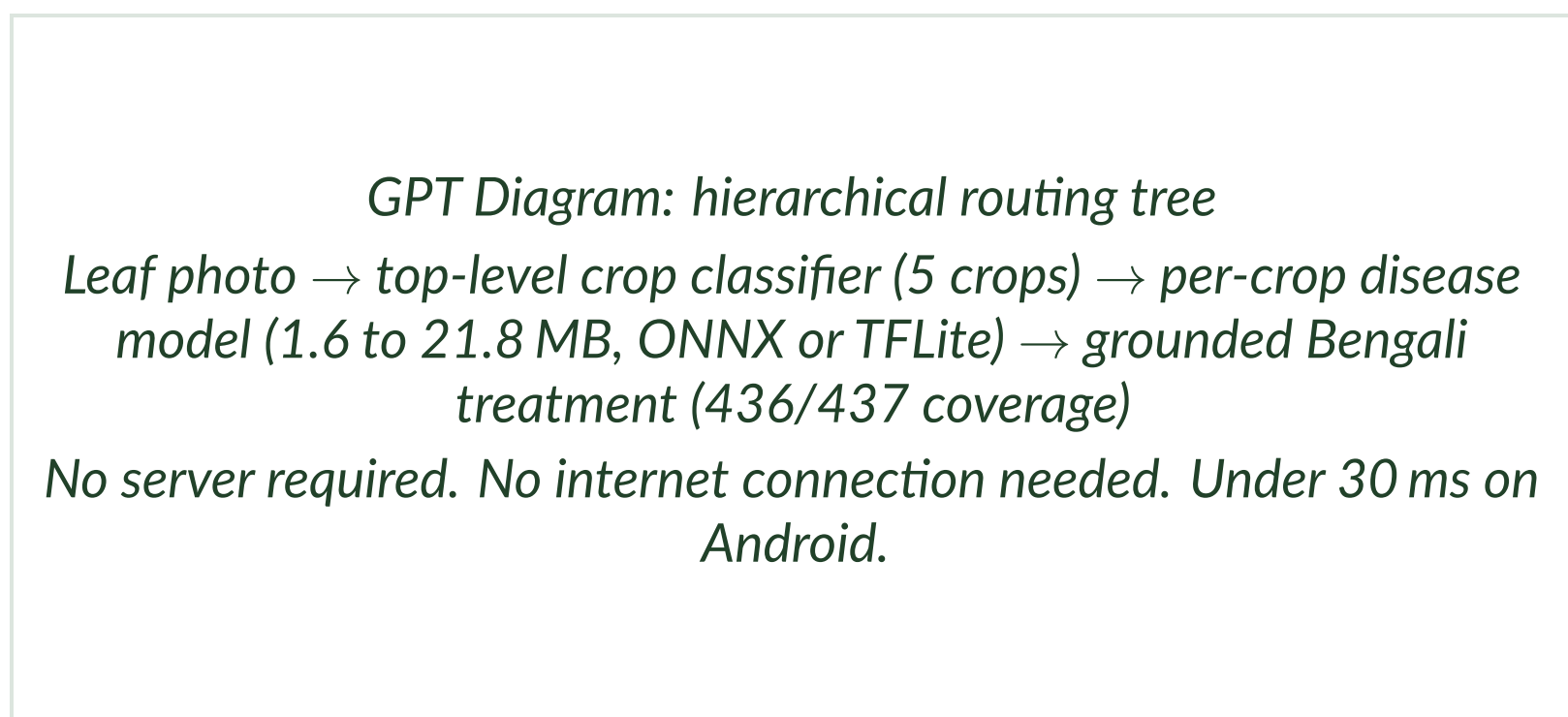
Why Each Stage Exists: Failure Before Fix

What we measured	What we built in response
LLMs produce wrong chemical dosages at 4.1–7.0% of queries even when the correct answer is given to them directly	Safety check runs before retrieval, plus a 14-field verifier that refuses to generate if evidence is absent
Dense retrieval scores R@10 of 0.970 on formal queries but drops to 0.093 on the same questions in colloquial farmer language	Hybrid BM25 + dense retrieval fused with Reciprocal Rank Fusion
92% of technical disease terms in official documents do not appear in any form in farmer queries; a farmer writes পাতা হলুদ হয়ে যাচ্ছে where the manual says “Yellow Mosaic Virus”	Retrieval is register-aware across 6 dialects
The national Krishi Call Center handles 92,094 calls in one year with a fixed human team	AI handles routine queries automatically; agents receive only high-risk cases with a full audit trail

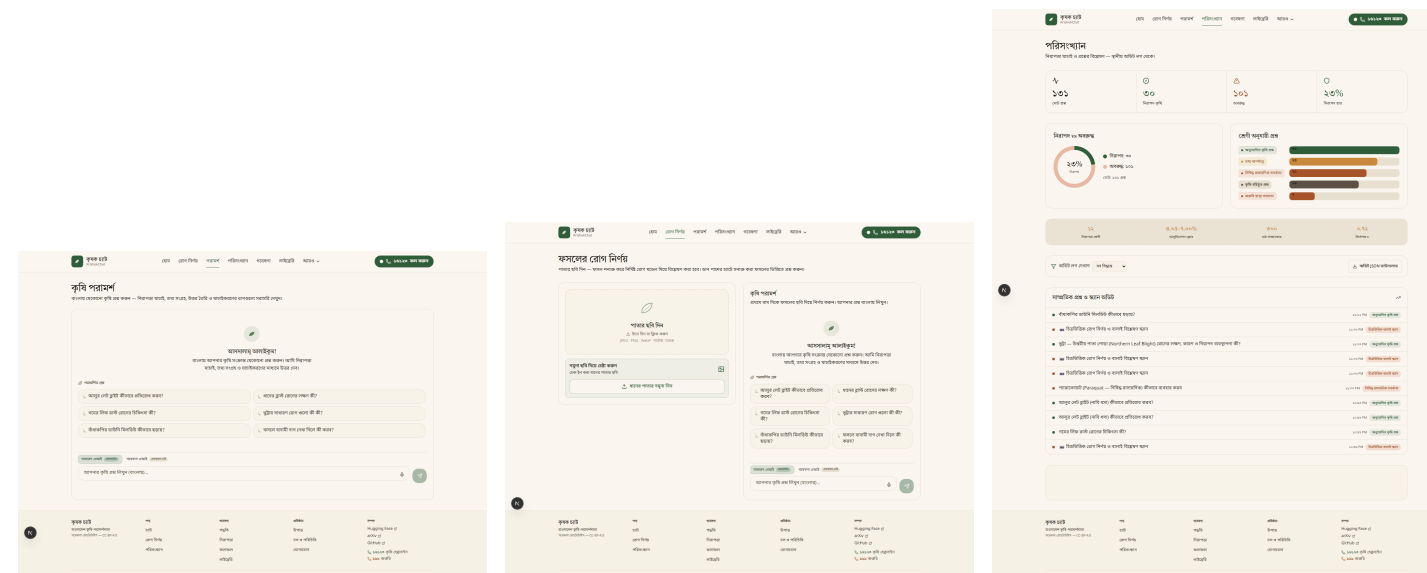
System Architecture: 4-Stage Safety Pipeline



On-Device Disease Classification and Routing



The Product: Krishi Patrak (কৃষি পত্রক)



Safety refusal → 16123

Photo → diagnosis

Audit dashboard

A Bengali-first web app and Android app. Every query includes a live agent trace showing the retrieval and verification steps. Supports voice input, a high-contrast sunlight mode, and works with no internet connection.

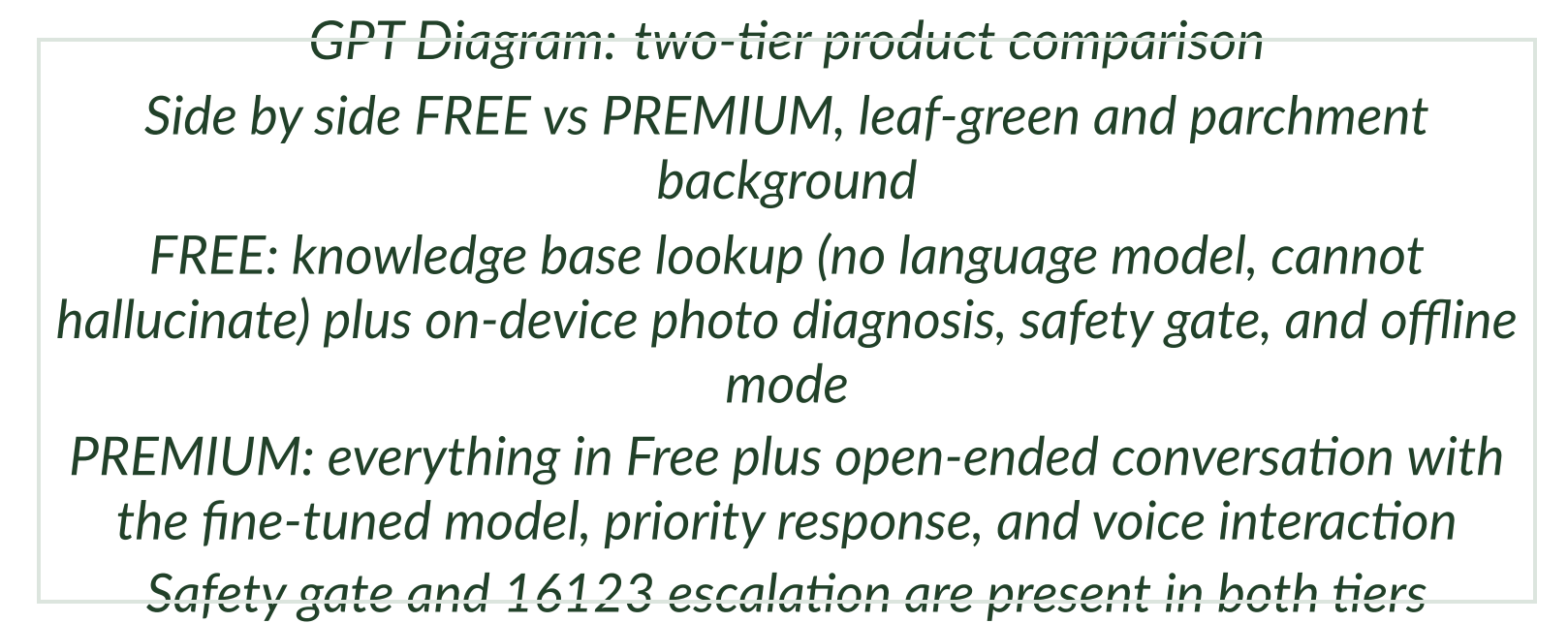
Market Readiness

What is working now. Full-stack web and Android app, on-device crop disease models, live safety pipeline with an audit dashboard, and a 437-entry verified treatment library tested end-to-end.

The offline case. 73.4% of Bangladeshi households own a smartphone, but only 43.6% of rural residents have internet access. A farming advisory that requires connectivity reaches less than half the intended audience.

Unit economics. Each AI answer costs roughly 2 paisa on a shared API. At scale, a single GPU server at \$497/month handles around 13 million answers per month.

Business Model: Farmers Use It Free; Institutions Pay



B2G. The system acts as an AI front desk for the Krishi Call Center. Routine queries are answered automatically. High-risk cases go to a human agent with a complete audit trail that satisfies DAE and a2i reporting requirements.

B2B. An analytics dashboard gives agro-dealers, seed companies, and NGOs district-level data on what farmers are asking and which crop diseases are rising in a given season.

Data licensing. The 85,979-instance benchmark and 722-image soil dataset are CC-BY-4.0 for academic use and available under a commercial licence for agri-fintech and crop insurance applications.

Papers, Code and Data



Dataset (Hugging Face)



Code (GitHub)

Full references listed in the footer below.

[1] KrishokChat: A Provenance-Traceable Multi-Task Bengali Agricultural Benchmark with Safety-Critical Chemical Advisory. *EACL 2026, under review.*

[2] AgriTrust: KG-Grounded Register-Aware Retrieval for Bengali Agricultural Advisory. *SIGIR-AP 2026, under review.*

[3] AgriVision BD: On-Device Crop Disease Classification. *Technical Report, 2026.*